Systems Thinking and Design

Phase 1 Report — The Breakfast Paradox at IIIT Hyderabad

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Contents

1 Introduction

This report consolidates the Phase 1 submission for the *Systems Thinking and Design* capstone project, addressing the "Breakfast Paradox" at IIIT Hyderabad: a large number of students register for mess breakfast, do not attend, and are still billed for the meal, while the kitchen still prepares food against the registration count.

Phase 1 of the course's Generic Systems Thinking & Design Project Framework asks for five deliverables, each corresponding to one activity:

1. System Context Brief
2. Stakeholder Map
3. Stakeholder Power–Interest / Leverage Map
4. Current-State Process Map / Process Trace
5. System Timeline / Behaviour-Over-Time Map

Deliverables 1 and 2 are complete, built from directly confirmed facts and a first-principles stakeholder-discovery pass. Deliverables 3, 4, and 5 are presented here as validated drafts: everything that could be established from existing confirmed facts and structural reasoning has been built out, but each contains hypotheses that are explicitly flagged and are being tested against real interviews with students, mess staff, and mess governance as fieldwork completes. Sections 4 and 5 in particular will be updated once interview data replaces the flagged hypotheses — these sections are written so that each hypothesis can be corrected in place without restructuring the document.

Throughout this report, three tags are used consistently to mark the confidence level behind a claim: [**Confirmed**] for a directly confirmed fact, [**Hypothesis**] for a plausible but unconfirmed inference, and [**Open question**] for a question that remains genuinely open.

2 Task 1: System Context Brief

Status: Complete

2.1 System Boundary

The system under study is the breakfast meal registration, billing, and service process operated across the four residential dining messes at IIIT Hyderabad — Kadamba, Palash, Bakul, and Yuktāhār — together with the digital registration and billing platform (`dining.iiit.ac.in`) that governs access to all four. The boundary includes the interface between this system and the academic class schedule, since the two operate on overlapping but independently controlled timings.

On-campus alternatives that compete for the same demand — Vindhya Canteen, the juice canteens, and third-party food delivery services — sit at the edge of this boundary. They are not part of the mess system's own operation, but they materially affect it: a student who does not eat at a mess draws instead on one of these.

Lunch and dinner registration at the same four messes fall outside this boundary. The system described here is scoped to the breakfast meal, though it runs on the same underlying platform and the same rules as the other two meals.

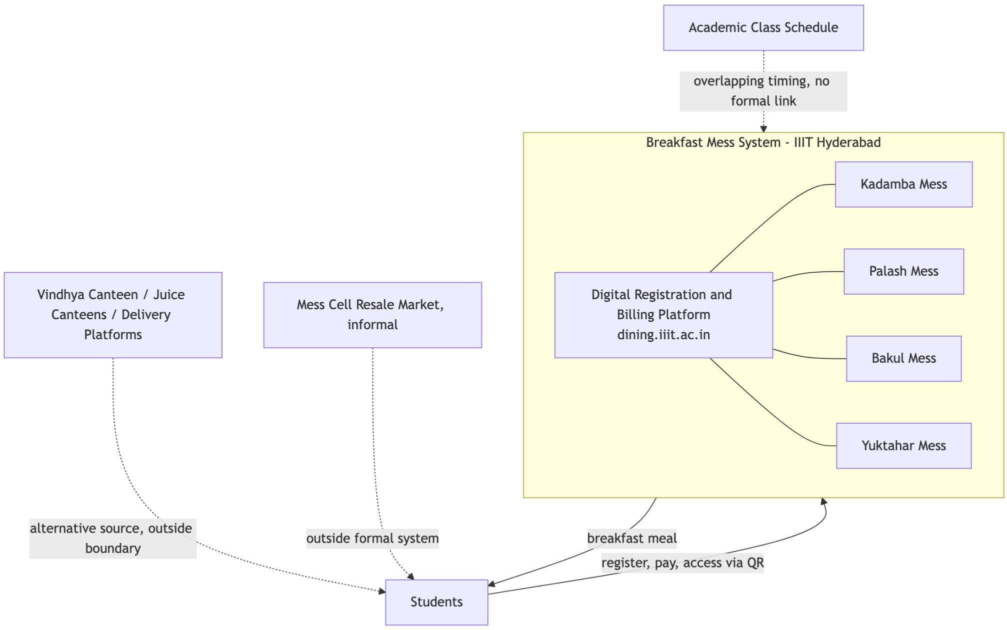


Figure 1: System boundary of the breakfast mess system.

2.2 Context

The breakfast system serves the entire resident student population of the institute across four messes, operating daily from 7:30 AM to 9:30 AM. Total confirmed breakfast capacity across the four messes is approximately 2,755 registrations per day: Kadamba (700 vegetarian, 600 non-vegetarian), Bakul (350 vegetarian, 350 non-vegetarian), Palāsh (400), and Yuktāhār (340 regular, 15 Jain).

Structurally, the system combines centralized demand management with decentralized supply. A single digital platform governs registration, billing, and access across all four messes, while food production itself is distributed and non-uniform, organized around cuisine rather than being interchangeable outlets:

- **Kadamba** — South Indian food, vegetarian and non-vegetarian; own kitchen and vendor.
- **Palash** — North Indian vegetarian food only; its kitchen cannot produce non-vegetarian food at all (a fixed physical limitation, not a policy choice).
- **Bakul** — North Indian food, vegetarian and non-vegetarian: its vegetarian food is cooked at Palash and transported to Bakul, while its non-vegetarian food is prepared on-site.
- **Yuktāhār** — Jain-oriented food, with a separate Jain registration variant, operating independently.

Food procurement is handled separately by each mess's own vendor, not centrally. Four different cuisines, kitchens, and procurement arrangements are unified only at the registration, billing, and access layer — the mess system is not one kitchen serving one population through one process, but a coordinated platform sitting over four structurally different operations.

The system operates inside, and is shaped by, several adjacent systems it does not control:

- **Academic administration and class schedule.** Classes begin at 8:30 AM and run until 6:40 PM, with a lunch break from 1:00 PM to 2:00 PM. There is no formal channel of coordination between the mess system and the academic calendar; the two schedules are set independently by separate authorities.
- **Warden, Mess Committee, and Mess Office governance.** Mess governance is split across three actors. The *Mess Committee* is a policy body, chaired by a faculty member, that sets the menu and mess-specific policy, incorporating student input approximately one month ahead of service. The *Mess Office* is the operational arm: it places food orders with each vendor on a four-day lead time, administers the registration platform, and issues operational policy changes — a policy change tightening cancellation rules was issued by the Mess Office specifically to reduce staff workload. The *Warden* holds authority over vendor management and structural changes to the mess; the working relationship between the Warden and the Mess Office is not established.
- **An informal peer-to-peer resale market ("Mess Cell").** A WhatsApp group has formed in which students who hold a breakfast registration they will not use offer it for resale to other students at a negotiated price below the registered rate. This market operates entirely outside the mess system's own registration and billing infrastructure, transacted through the same personal QR-based access credential the formal system issues.
- **The wider campus food ecosystem.** Vindhya Canteen (open Monday to Saturday, from 10:00 AM), the juice canteens (open mornings), and delivery platforms (accepting orders until 11:00 PM, with delivery personnel required to remain outside the campus gate) all draw on the same student population and stand as alternatives to mess breakfast.

Billing operates on a monthly cycle tied to registration, not attendance: a student is charged for every meal they are registered for, regardless of whether they eat it, and may cancel a registration in advance, subject to a limit of five cancellations per meal type per month. Unregistered ("walk-in") access to a meal is priced materially higher than registered access to the same meal.

2.3 Purpose and Intended Outcomes

Purpose. The breakfast mess system exists to provide the resident student population with reliable, adequate, good-quality food each morning, at a predictable and affordable cost, accommodating dietary requirements — vegetarian, non-vegetarian, and Jain — in a manner that supports student health, energy, and academic performance.

Intended outcomes.

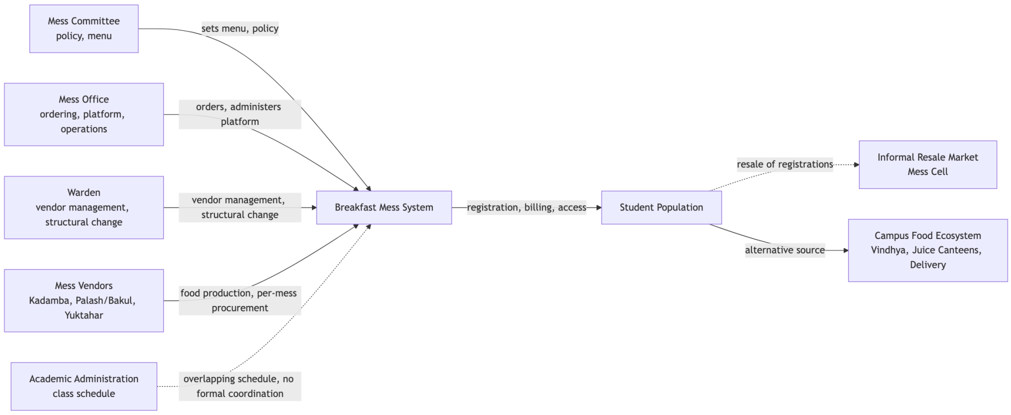


Figure 2: Context map showing the mess system and the adjacent systems that shape it.

- Every registered student obtains a breakfast meal matching their dietary requirement within the operating window.
- Food quality and preparation quantity remain consistent with actual demand, minimizing both shortage and waste.
- The cost structure remains affordable and predictable for students across a full semester.
- Access to breakfast does not depend on a student's ability to navigate informal workarounds; the formal system functions as the primary and sufficient channel on its own.
- The system remains responsive to changes in student needs and circumstances — schedule conflicts, dietary changes, feedback — through its own governance structure.

2.4 Visible Symptoms and Underlying Issues

Visible Symptom	**Underlying Issue**
A student registers for breakfast, does not attend, and is still charged the full registered price.	Billing is tied to registration, not attendance. The cost is incurred at the point of registering, independent of whether the meal is ultimately consumed.
The registration platform imposes a cap of five cancellations per meal type per month.	Uncancelled, unattended registrations occur often enough that an explicit limit was necessary to manage them — the cap is itself evidence of the scale of the pattern it constrains.
A "Skip Meal" function lets a student notify the kitchen in advance of non-attendance, but this does not reduce or refund the charge.	The mechanism addresses only the kitchen's production planning, not the student's cost. It returns zero value to the student for the same situation a cancellation — capped at five per month — would address instead.

Visible Symptom	Underlying Issue
An unofficial, student-run resale market ("Mess Cell") has formed, where students sell registrations they will not use to other students.	Reselling returns partial or full value to the student, while Skip Meal returns none. For a student who knows in advance they will not attend, resale is a materially better option than the system's own built-in mechanism.
Registered-but-unclaimed meals persist despite both a cancellation option and a resale market being available.	Cancellation and resale both require advance knowledge of non-attendance. Last-minute non-attendance — oversleeping, running out of time, a change of plans — has no mechanism available to it at all; this is the category that converts into prepared, uneaten food with no value returned to anyone.
Walk-in (unregistered) access to a meal costs substantially more than the same meal accessed through a prior registration.	The pricing structure creates a standing incentive to register in advance even when attendance is uncertain, since deciding same-day costs more. Registration counts are inflated by this incentive independent of actual intention to attend.
Registered breakfast attendance is markedly lower, relative to capacity, at every mess serving North Indian food (Palash, Bakul) than at Kadamba, the South Indian mess — Kadamba runs 51–79% uptake against 11–36.5% at the North Indian lines.	The gap tracks cuisine, not any single mess's individual circumstances — it holds consistently across three independently operated messes sharing that one variable. Bakul's status as a newer mess remains a contributing factor, but the cuisine pattern is the stronger and better-evidenced explanation.
Neither students nor mess-side staff describe a specific example of mess feedback leading to a change in policy, hours, or operations.	Three real channels carry student input into the system — an in-app per-meal rating, a public email address, and coverage in *Ping*, the student publication. None of the three has a confirmed instance of producing a policy change.
Breakfast hours (7:30–9:30 AM) overlap with the start of the academic day (8:30 AM), while the two schedules are set by entirely separate authorities.	Neither side has authority over the other's schedule, and no coordination mechanism links the two. A timing conflict between them can persist indefinitely without either side being positioned to resolve it unilaterally.
The registration platform includes a random-allocation mechanism, evidenced by a settings option to notify a student when they have been randomly allocated a meal.	Some registrations are generated by the system itself rather than by active student choice. Recorded registration counts do not uniformly represent expressed intent to attend.
A portion of students report skipping breakfast for reasons unrelated to the mess system's own operation — personal dietary practice, sleep schedule, established habit.	Breakfast non-attendance is not driven by a single cause; some non-attendance would persist regardless of any change to mess hours, pricing, or process.

Visible Symptom	Underlying Issue
Students report that companionship affects whether they attend breakfast — attendance is more likely when a roommate or friend is also going.	Breakfast attendance, for at least part of the student population, is a socially coordinated decision rather than a purely individual one.

2.5 Key Actors, Institutions, Processes, Resources and Constraints

Actors

- **Students** — registrants and consumers of the meal; attendance and dietary preference vary across the population.
- **Mess Committee** — a faculty-chaired policy body that sets the menu and mess-specific policy.
- **Mess Office** — the operational and administrative arm; places food orders, administers the registration platform, issues operational policy changes.
- **Warden** — holds authority over vendor management and structural changes to the mess.
- **Per-mess vendors** — Kadamba's vendor (South Indian); the shared Palash/Bakul vendor (North Indian); Yuktāhār's vendor (Jain-oriented).
- **Mess serving and cleaning staff** — also consume mess food after student service hours conclude.
- **Academic administration** — sets the class schedule; holds no authority over mess operations.
- **Students participating in the Mess Cell resale market** — an unofficial secondary role occupied by a subset of registered students.
- **Guests** — parents, visiting relatives, and non-student campus affiliates who pay the vendor directly for access.

Institutions

The Mess Committee (policy), the Mess Office (operations), and the Office of the Warden (vendor management and structural change) are three separate governance actors holding different parts of mess governance. The academic administration/timetable authority operates independently with no formal link to any of the three mess-governance actors. The four mess vendors each handle their own procurement and cooking, operating as independent entities under one shared registration platform.

Processes

- **Registration** — a student selects a mess and meal date through the digital platform in advance of service.
- **Billing** — charges are applied monthly, calculated from registration count, independent of attendance.
- **Cancellation** — a registered meal may be cancelled in advance, up to five times per meal type per month, removing the associated charge.

- **Skip Meal** — a registered student may flag in advance that they will not attend; this notifies the kitchen for production planning but does not affect billing.
- **Access** — a student presents a personal QR code at the mess counter to receive a plate matching their registered dietary category.
- **Walk-in access** — a student without a registration may pay the vendor directly at a higher rate, subject to a separate, smaller capacity allocation.
- **Resale** — outside the formal platform, a registered student may transfer their registration to another student through the Mess Cell WhatsApp group, in exchange for a negotiated payment.
- **Menu setting** — the Mess Committee finalizes the menu approximately one month ahead of service, incorporating student input.
- **Food ordering** — the Mess Office places an order with each vendor four days ahead of service; vendors prepare food against that four-day-old figure, not against same-day registration or cancellation activity.
- **Feedback intake** — student input reaches the system through an in-app per-meal rating, a public email address, and coverage in *Ping*; none of the three has a confirmed record of producing a resulting policy change.

Resources

The digital registration and billing platform (`dining.iiit.ac.in`); four physical mess facilities of differing capacity and origin — Kadamba (purpose-built, largest capacity), Bakul (a converted warehouse), and Palash and Yuktāhār as separate facilities; per-mess food production capacity, distributed unevenly by cuisine; and the personal QR access credential issued to each student.

Constraints

- Billing is registration-based, not attendance-based, and cancellations are capped at five per meal type per month.
- Walk-in pricing is set materially higher than registered pricing.
- Vegetarian and non-vegetarian food is segregated by registration type at the point of service.
- Palash's kitchen cannot produce non-vegetarian food — a fixed physical limitation, not a policy choice.
- Mess operating hours and the academic class schedule are set independently by separate authorities, with no formal coordination mechanism between them.
- Governance authority over mess operations is split across the Mess Committee, the Mess Office, and the Warden; the academic administration has no formal role in mess-related decisions.
- Vendors order and prepare food against a four-day-old registration figure; nothing in the system lets same-day registration, cancellation, or Skip Meal activity change what is actually cooked.

2.6 Initial Problem Framing

Many students register for breakfast and do not eat it. The meal is still cooked, and the student is still charged.

The system gives students two ways to avoid this: cancel the registration, or mark it as skipped. Skip Meal returns no money to the student — only cancellation does, and only five times a month per meal. Both require the student to decide in advance. A student who decides at the last minute has no option at all, and that meal is simply wasted. Students have built their own workaround outside the system — reselling registrations to each other — because it pays back more than the system's own tools do.

> **The problem:** registration does not reliably reflect who is actually going to eat. Why not, and what in the system is causing that gap?

3 Task 2: Stakeholder Map

Status: Complete

3.1 Use Case

The breakfast mess system at IIIT Hyderabad registers, bills, and serves breakfast to the resident student population across four messes — Kadamba, Palash, Bakul, and Yuktāhār — through a single digital platform. Registration counts, not attendance, drive billing and food procurement. This map identifies every actor, institution, and informal structure that participates in, influences, benefits from, or is affected by this system, and the relationships, dependencies, and conflicts between them.

Twenty-seven candidate actors were identified across seventeen analytical lenses (student/user, social, food production, governance, money, technology, academic, sleep/time-use, substitute systems, campus infrastructure, waste, health, policy, informal systems, absent/sleeping stakeholders, temporal, and intervention). Seventeen were included in the final map; four were treated as minor rather than included in full; three were considered and excluded entirely (a campus health/wellbeing office, senior-to-junior norm transmission, and a parental financial relationship, since mess fees are bundled into hostel fees).

3.2 Primary, Secondary, and Indirect Stakeholders

Primary stakeholders — direct, daily interaction with the system:

Stakeholder	Type
Students	Human group — registrant, billed party, consumer
Vendor — South Indian (Kadamba)	Commercial
Vendor — North Indian (Palash veg-only + Bakul veg/non-veg)	Commercial
Vendor — Jain (Yuktāhār)	Commercial
Mess serving/counter staff	Human group
Kitchen/cooking staff	Human group

Secondary stakeholders — govern, execute, or manage the system without daily direct interaction:

Stakeholder	Type
Mess Committee	Institution — policy
Mess Office	Institution — operations
Warden	Individual role — vendor management, structural change

Indirect stakeholders — affected by or connected to the system without direct engagement, including latent ("sleeping") actors currently inert but holding potential leverage:

Stakeholder	Type
Academic administration / timetable authority	Institution — latent
Facilities/Estate team	Institution — latent

Stakeholder	Type
Portal/IT system owner	Institution or individual — latent
Mess Cell WhatsApp community	Informal group
Unofficial mess-tool developers	Individual(s) — latent
Ping (student publication)	Institution — latent
Vindhya Canteen and juice canteens	Commercial — substitute
Delivery platforms and riders	Commercial — substitute

3.3 Stakeholder Interests, Needs, and Expectations

Stakeholder	Interest	Need	Expectation
Students	Convenient, affordable, timely food; not paying twice for a missed meal	A functioning registration/billing system, predictable hours, food matching dietary registration	Registering reliably results in being fed; being charged regardless of attendance is tolerated, not agreed
Vendor — Kadamba (South Indian)	Contract renewal, predictable order volume, margin	An accurate registration forecast, currently four days ahead	The four-day forecast approximates real demand
Vendor — Palash + Bakul (North Indian)	Contract renewal, predictable order volume, margin	Same four-day lead time; Palash cannot produce non-veg food, a fixed limitation	Demand for the veg line cooked at Palash and transported to Bakul stays predictable
Vendor — Yuktāhār (Jain)	Contract renewal, predictable order volume, margin	Same four-day lead time; relationship to other vendors unconfirmed	Same as other vendors
Serving and kitchen staff	Manageable preparation and service load, predictable headcounts	Sufficient lead time and an accurate registration count	Leftover food is theirs to consume after service hours — an informal norm
Mess Committee	Balancing student satisfaction against operational feasibility	A functioning process for gathering and interpreting student input	The existing menu-input process constitutes sufficient engagement on mess matters generally
Mess Office	Reduced operational workload — its stated motive for its one confirmed policy change	An accurate same-day picture of intent, currently unavailable at less than four days' notice	Registration counts are a workable proxy for actual attendance
Warden	Stable vendor relationships, orderly mess infrastructure	Clear division of responsibility with the Mess Office	Vendor management and structural authority remain within the Warden's own lane
Academic administration	Its own domain — class scheduling — run without external interference	No stated need connects it to mess operations	No expectation of any relationship to mess governance

Stakeholder	Interest	Need	Expectation
Facilities/Estate team	Physical infrastructure upkeep	Authorization and budget for mess-facility work	Not established — identity of this actor is itself unconfirmed
Portal/IT system owner	System uptime and reliability of the registration interface	Not established	Not established — no source names this actor
Mess Cell WhatsApp community	Recovering value from a registration that will go unused	A way to transfer a registration since Skip Meal returns no money	Buyers and sellers can find each other informally and transact on trust
Unofficial mess-tool developers	Not established beyond building the tool itself	Not established	Described as built "for students of IIIT Hyderabad"
Ping (student publication)	Reporting on mess issues as a matter of student interest	Access to information worth publishing	Not established whether coverage produces institutional response
Vindhya Canteen and juice canteens	Capturing demand when mess breakfast is skipped or unavailable	Steady walk-in customer flow	Not established
Delivery platforms and riders	Capturing demand when mess breakfast is skipped or unavailable	Gate access for delivery, currently restricted	Not established

3.4 Formal and Informal Roles

Formal roles attach to every institutional and commercial actor: Students (as registrant), Mess Committee, Mess Office, Warden, the three vendors, serving and kitchen staff, Vindhya Canteen and juice canteens, and delivery platforms all hold a formal, rule-defined position in the system.

Informal roles attach to the Mess Cell WhatsApp community, Ping, and the unofficial mess-tool developers — none of these has any official standing in the system, yet each performs a real function (resale market, public-pressure channel, and an alternative registration interface, respectively).

Roles that are not separate stakeholders. Several categories that might appear as distinct actors are, on inspection, behavioral states or temporary roles occupied by the single actor Students, not separate stakeholders in their own right: attending regularly, skipping regularly, having an early class on a given day, buying or selling on Mess Cell, and using an unofficial registration tool are all states a single student may move through within the same week. Peer influence between students is an internal relationship on the Students node, not a separate "Friends" or "Roommates" stakeholder. Guest diners are the one genuine exception: they hold a materially different formal relationship, but their volume is too marginal to warrant inclusion as a full stakeholder.

3.5 Stakeholder Maps

3.6 Relationships, Dependencies, and Conflicts

Relationship matrix

Cells marked [**Hypothesis**] have not been independently confirmed.

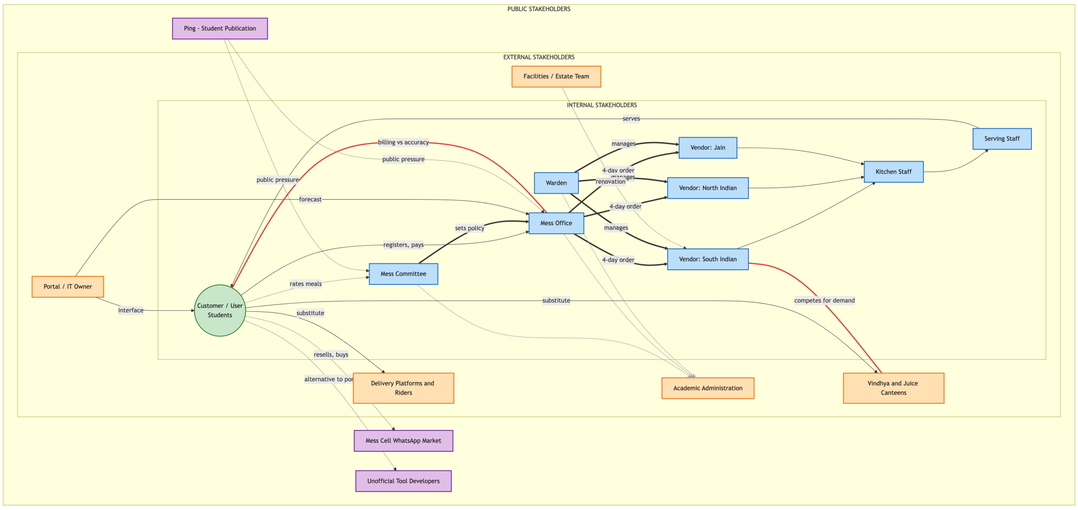


Figure 3: Stakeholder map, placed by the standard four-ring model (customer/user, internal, external, public).

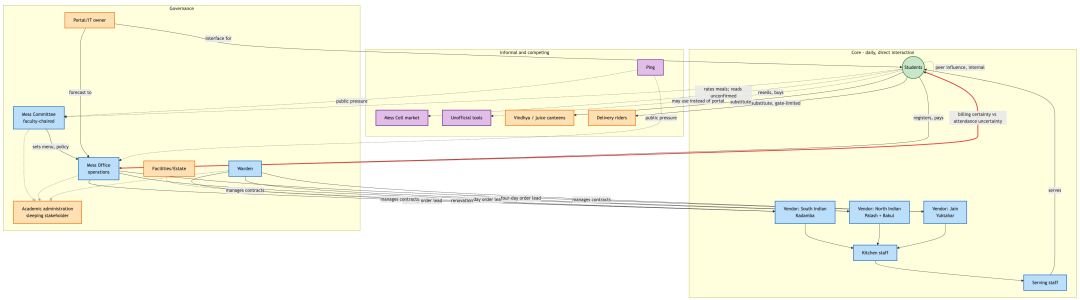


Figure 4: Relationship network — the same seventeen stakeholders laid out by functional cluster (core daily operation, governance, informal/competing).

	Students	**Mess Committee**	**Mess Office**	**Warden**	**Vendors**
Students	—	Feedback (rating system confirmed; whether read is unconfirmed)	Registration, payment	—	Payment, service
Mess Committee	Authority (menu, feedback)	—	Authority (sets policy)	[**Hypothesis**] overlap	Authority (menu)
Mess Office	Billing; dependent on for order accuracy	Reports to ([**Hypothesis**])	—	[**Hypothesis**] overlap	Authority, dependency (ordering)
Warden	—	[**Hypothesis**]	[**Hypothesis**]	—	Authority (contracts)
Vendors	Payment, food/material	Authority over	Authority over, dependency	Contract authority	—

Academic administration has no confirmed relationship of any kind to any of the three governance actors — a genuine structural gap, not a modeling omission, and the single most consequential finding in this map.

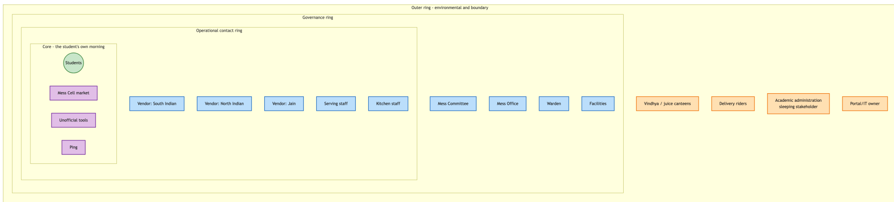


Figure 5: Onion proximity map — stakeholders clustered by closeness to a student's actual morning rather than by relationship type. Academic administration sits at the outermost ring with no adjacent inward relationship at any layer, while Mess Cell, unofficial tools, and Ping sit at the innermost ring alongside the students themselves.

Dependencies

Students depend on vendors for food quality and timing; vendors do not depend on any individual student — an asymmetric dependency consistent with a walk-in pricing structure that requires no negotiation. Vendors depend on the Mess Office for a registration forecast fixed four days ahead of service; the Mess Office depends on students to register accurately, but nothing in the billing structure requires accuracy, since a student is charged identically whether or not they attend — the weakest link in this dependency chain is also the one with no incentive attached to it.

The Mess Committee depends on the Mess Office to execute policy; the Mess Office's own stated motivation for its one confirmed policy change (reducing workload) did not visibly originate from or require Mess Committee direction, suggesting operational pressure may drive policy from below at least as often as governance drives it from above.

No source describes a direct relationship between vendors and students — all communication routes through the Mess Office or the portal. No source describes the Portal/IT owner having any direct relationship with the Mess Committee, despite the Committee setting policy this system presumably enforces technically.

Conflicts

Conflict	Side A	Side B
Billing certainty vs. attendance uncertainty	Mess Office and vendors need a stable number to order against, fixed four days ahead	Students face genuinely uncertain day-to-day attendance, with no incentive built into billing to report it accurately
Flexibility vs. workload	Students and the Mess Cell market want more cancellation and resale flexibility	The Mess Office's one confirmed policy change tightened rules specifically to reduce staff workload
Convenience vs. waste	Students want flexible registration	The Mess Office wants predictable, low-workload ordering
Cuisine demand vs. fixed procurement	Registered uptake is markedly lower at every North Indian line (Palash, Bakul) than the South Indian line (Kadamba)	The four cuisine lines are fixed by vendor contract, not adjusted in response to demand

Conflict	Side A	Side B
Academic schedule vs. meal schedule	Academic administration owns class timing and holds no stake in mess outcomes	Students bear the actual cost of the 8:30 AM overlap with breakfast hours

3.7 Findings

Twenty-seven candidate stakeholders were narrowed to seventeen through direct evidence rather than assumption. The Mess Office is a distinct actor from the Mess Committee, separating "who decides policy" from "who executes it" — every earlier account of this system's governance, including this team's own first drafts, had collapsed the two into one.

Academic administration holds the single lever most likely to materially affect the core problem — the overlap between breakfast hours and the 8:30 AM class start — and has no relationship of any kind, formal or informal, to any of the three mess-governance actors. This is the clearest sleeping stakeholder in the map: total authority over the one variable in conflict, zero engagement with the system it conflicts with.

Three real feedback channels exist — an in-app per-meal rating, a public email address, and coverage in *Ping* — and none has a confirmed instance of producing a policy change. Inputs exist; a confirmed output does not.

Registered attendance is markedly lower, relative to capacity, at every mess serving North Indian food than at the South Indian mess, a pattern consistent across three independently operated messes, making cuisine preference a stronger candidate explanation for the uptake gap than any single mess's individual circumstances.

The Mess Committee's faculty chair is a second, more subtle sleeping stakeholder: an individual with cross-institutional standing that no student-only channel possesses, and the most structurally plausible bridge to Academic administration available, with no evidence that bridge has ever been used.

Two structurally hidden actors carry outsized influence without being visible to any single respondent: the Portal/IT system owner, interacted with constantly and named by no one, and the four-day procurement lead time itself — not a person, but a fixed constraint that caps how responsive the entire system can be to any same-day signal, including Skip Meal.

4 Task 3: Stakeholder Power–Interest / Leverage Map

Status: Draft — built from confirmed context-brief and stakeholder-map facts; to be finalized once governance-side (Role D) interview data lands

4.1 Power Analysis Brief

Power is defined as decision-making authority plus resource control. Interest is how much the outcome affects the actor, or how much attention they actually pay it — these are not always the same thing, as the Academic Office row below shows.

Actor	Power	Interest	Quadrant
Mess Committee	High — sets menu/rules	High	Manage Closely [**Confirmed**]
Warden	High — vendor management, structural changes	Medium-High	Manage Closely [**Confirmed**]
Per-mess vendors/catering providers	Medium-High — controls supply volume/quality/cost	Medium	Manage Closely / Keep Satisfied [**Hypothesis**]
Academic Office / timetable authority	High — controls the one lever (class start time) that could resolve the core paradox	Very Low — no mandate/attention on mess matters	Nominally "Keep Satisfied", functionally disengaged — a **sleeping power** [**Hypothesis**]
Students	Low — no formal decision authority, "input" only	Very High — bears cost and experience directly	Keep Informed — a textbook mismanaged quadrant [**Confirmed**]/[**Hypothesis**]
Mess serving/kitchen/cooking staff	Low formal power — execute decisions	High — workload, waste, day-to-day friction	Keep Informed — an underused source of operational truth [**Hypothesis**]
Mess Cell WhatsApp community	None formal, but high de facto structural influence (absorbs waste/no-show pressure the official system doesn't handle)	High	Monitor formally / arguably deserves Manage Closely once its real scale is known [**Hypothesis**]
Vindhya/juice canteens, delivery riders	Low power over the mess system itself	Medium — business interest in capturing skippers	Monitor [**Confirmed**]
Friends/Roommates	Low power at system level	High interest at the individual-decision level, not systemic	Out of formal power–interest scope — relevant to the Process Trace instead [**Hypothesis**]

4.2 Quadrant Chart

4.3 Strategic Findings

1. **Academic Office is a "sleeping power."** It holds the single lever most likely to resolve the core paradox — shifting the 8:30 AM class start, or even just the first class of the week — but has no interest or mandate that would make it act. This is a structural / mental-model finding to carry into the Phase 2 Iceberg analysis, not just a gap in the map.

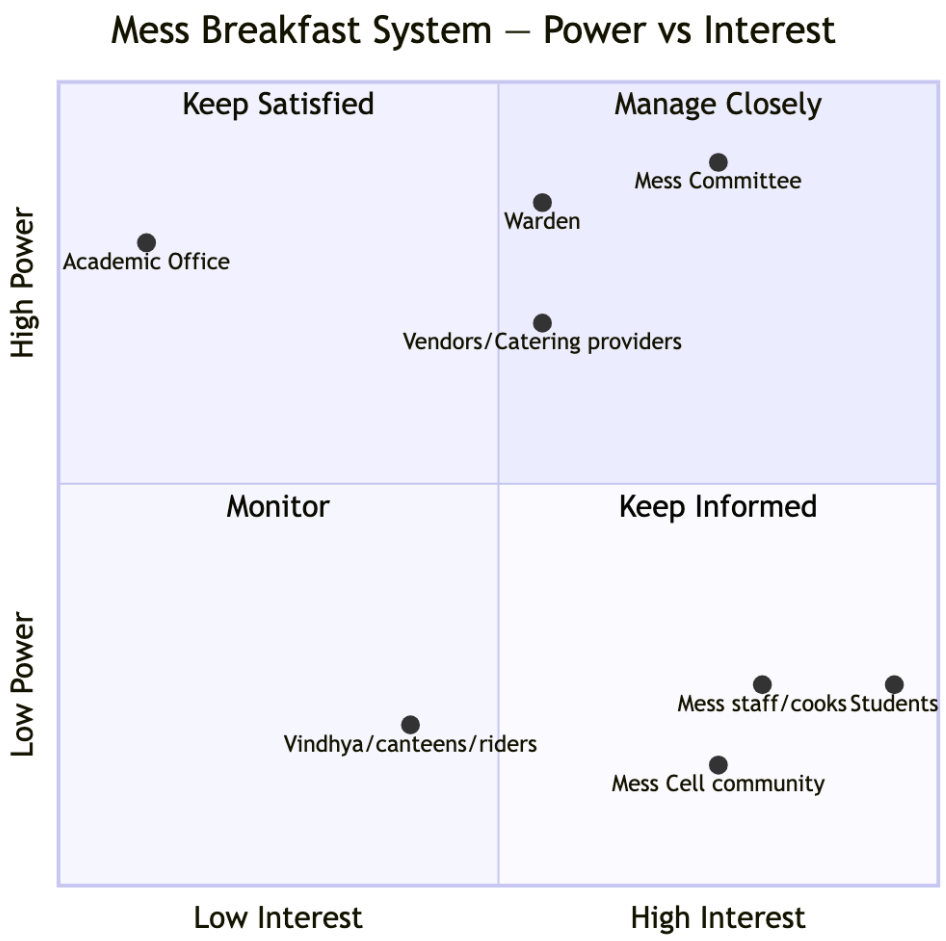


Figure 6: Power–Interest quadrant chart for the mess breakfast system.

2. **Mess staff are high-interest, near-zero-formal-power**, and their operational knowledge (waste volume, real headcounts, cost/workload pressure) does not appear to reach decision-makers through any formal channel. The mess-staff (Role C) interview is as much a data point for this map as it is for the Process Trace.

4.4 Open Items — to be closed by interviews

- Confirm the real scale/frequency of Mess Cell activity — would move it from "Monitor" toward "Manage Closely" if larger than assumed.
- Confirm students' own sense of their interest/power (currently plotted from the context brief's structural facts, not from a student's own framing).
- Confirm the vendor vs. catering-provider power split once staff interviews clarify which messes use which model.

5 Task 4: Current-State Process Map / Process Trace

Status: Draft — awaiting real interview and observation data (Activity 4 explicitly requires "interviews and observations"); everything below is the best-effort scaffold to be corrected, not a result to cite

5.1 Why a Process Trace, Not Just a Stakeholder Map

The stakeholder map says *who* is involved. This section is supposed to say *what actually happens*, step by step, including the workarounds — per the framework's own phrasing, "how the system actually operates rather than how it is supposed to operate." The gap between those two is exactly where the Breakfast Paradox lives: the *designed* process is register → attend → eat; the *actual* process, per every account gathered so far, has several parallel exit ramps around "attend."

Confirmed context-brief facts describe billing and cancellation at the *monthly* level, but a plausible *weekly* registration habit may sit on top of the *daily* attendance decision — three different timescales that a single word, "process," can hide if traced as only one loop.

5.2 A. Weekly Cycle

[**Hypothesis**] Unconfirmed by any real respondent yet.

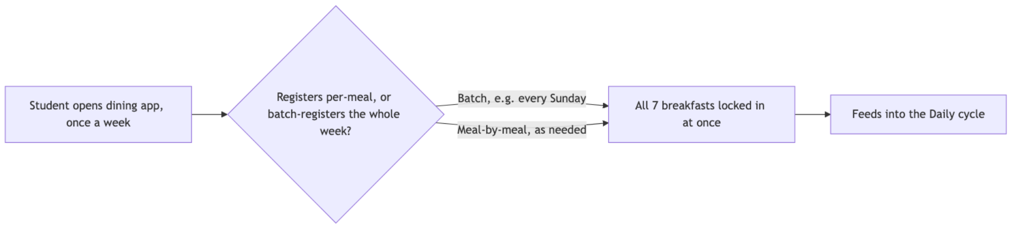


Figure 7: Weekly registration cycle (hypothesis).

If a batch-registration pattern (registering for the whole week at once, rather than meal-by-meal) is real and common, it mechanically decouples the registration decision from the morning-of attendance decision — a student who registers Sunday for Wednesday is not "deciding to eat Wednesday," they are deciding once for the whole week. This would explain a large share of the no-show volume without any single morning being anyone's fault, and is the single highest-value hypothesis to confirm through direct interview.

5.3 B. Daily Cycle (7:30–9:30 AM window)

5.4 C. Exception and Workaround Paths

5.5 Formal vs. Informal Rules Surfaced by This Trace

Step	Formal rule (as designed)	Informal reality (as it actually runs)
Registration	Per-meal opt-in via app	[**Hypothesis**] Possibly batched weekly, decoupling the decision from any single morning

Step	Formal rule (as designed)	Informal reality (as it actually runs)
No-show	Billed regardless [**Confirmed**]	A five-per-month cancellation cap exists specifically because uncancelled no-shows are common enough to need a limit — the rule is a symptom of the pattern, not just a control on it
Kitchen over-preparation	"Skip Meal" toggle exists to prevent it [**Confirmed**]	Toggle doesn't touch billing — solves the kitchen's problem, not the student's
Unused registration	No official resale/refund mechanism	Mess Cell WhatsApp group — fully informal, self-organized, price set peer-to-peer with no oversight [**Confirmed**] (exists), [**Hypothesis**] (scale)
Student feedback on rules	Mess Committee "takes student input" on menu [**Confirmed**]	No confirmed negotiation or escalation mechanism exists beyond an occasional feedback form — a real gap in the trace

5.6 Negotiation and Escalation Mechanisms

The framework asks this deliverable to identify "negotiation and escalation mechanisms." Current data has almost none to trace: the Warden handles vendor/structural changes, the Mess Committee sets the menu "with student input," and that is the extent of any confirmed channel. No respondent so far has described what a student actually does if they disagree with a mess decision, beyond vague memories of a petition or feedback form that nobody could confirm led anywhere. This absence is itself a process-trace finding, worth carrying into the Phase 2 Iceberg Structures layer as a real gap rather than an unanswered question.

5.7 What Is Confirmed vs. Hypothesized in This Section

- [**Confirmed**] **Fully confirmed by the context brief:** billing-by-registration, five-per-month cancellation cap, Skip Meal exists and does not affect billing, Mess Cell exists, walk-in markup, items running out does happen.
- [**Hypothesis**] **Hypothesized, needs a real interview:** batch weekly registration, a "wake-before-8:30 threshold" effect, a roommate-prompt mechanism, Skip Meal being forgotten rather than actively rejected, Mess Cell's real transaction scale.
- **Not yet traceable at all:** the mess-side process (kitchen prep timing, what staff actually do with unclaimed food) — entirely dependent on the mess-staff (Role C) interview; this section currently only traces the student-facing half of the process.

6 Task 5: System Timeline / Behaviour-Over-Time Map

Status: Draft — direct observation window has already passed for this cycle; this section substitutes existing screenshot data, confirmed context-brief facts, and retrospective recall pending interview confirmation

Two timescales matter here, and they are kept separate rather than collapsed into one chart.

6.1 A. Within-Window Pattern (7:30–9:30 AM, Half-Hour Scale)

[**Hypothesis**] Nothing in this table is measured; it is a plausible curve based on general dining/class-start dynamics, written down specifically so it can be *overwritten*, not just annotated, once interview data comes back.

Window	Hypothesized crowd level	Reasoning (to be tested)
7:30–8:00	Moderate	Early-class students and habitual early risers
8:00–8:30	Peak	Closest to the 8:30 class start — highest queue pressure
8:30–9:00	Dip	Anyone with an 8:30 class has already gone or given up
9:00–9:30	Small second bump	Late risers with no early class, before close

6.2 B. Across-Time Pattern (Week Scale)

Date	Meal	Type	Detail	Source
2026-09-03	Breakfast	Individual instance	Respondent's own Kadamba (Veg) registration, charged Rs. 48, not attended	Screenshot
2026-09-09	Breakfast	Mess-wide capacity	Kadamba (Veg) 551/700 registered	Screenshot
2026-09-09	Breakfast	Mess-wide capacity	Bakul (Veg) 108/350 registered	Screenshot
2026-09-09	Breakfast	Mess-wide capacity	Bakul (Non-Veg) 37/350 registered	Screenshot
2026-09-10	Lunch	Mess-wide capacity	Kadamba 722/1200 registered	Screenshot
2026-09-10	Lunch	Mess-wide capacity	Bakul 116/700 registered	Screenshot

[**Hypothesis**] Three scattered dates across two different meals is not enough to plot a real trend line — a smooth graph through these points would fabricate a precision that is not there. What this table does consistently support, on every date sampled so far, is a persistent Kadamba $\gg$ Bakul uptake gap, at both breakfast and lunch.

6.3 The Recurring Pattern and Unintended Consequence

This finding does not need a numeric chart to count as a Behaviour-Over-Time result, and is already fully supported by facts confirmed in Task 1:

1. **Trigger (a structure):** billing is by registration, not attendance.
2. **Recurring pattern:** students register and do not show up, often enough that a five-cancellations-per-month cap exists — a cap that only makes sense if uncancelled no-shows are common. [**Confirmed**]
3. **The system's own partial fix, and its unintended consequence:** the "Skip Meal" toggle exists so kitchens stop over-preparing for known no-shows — but it does not touch billing, so it solves the kitchen's waste problem without touching the student's cost problem. [**Confirmed**]
4. **Emergent adaptation:** the Mess Cell WhatsApp resale market appeared, unofficially, as a workaround the system's designers did not build for. [**Confirmed**]

This reinforcing-loop-shaped structure feeds directly into the Phase 2 Causal Loop Diagram deliverable.

6.4 Feedback Loops and Systems Archetypes

Quick Iceberg pass

- **Events:** the September 3 charged-not-attended instance; items running out near the end of service; Kadamba sitting near capacity while Bakul sits far under it.
- **Patterns:** registration consistently exceeds attendance (the cap exists because of this); Kadamba $\gg$ Bakul uptake on every date sampled, at both meals.
- **Structures:** billing-by-registration; the five-per-month cancellation cap; Skip Meal (a kitchen-only fix); the informal Mess Cell market; [**Hypothesis**] possible batch-weekly registration; no confirmed escalation channel.
- **Mental models (all [Hypothesis], none directly confirmed yet):** "I already paid for it, the money's gone either way"; "giving feedback doesn't really change anything"; "Kadamba is just the trusted default."

R1 — "Shifting the Burden" via Mess Cell

Reinforcing — sustains the dysfunction rather than growing it unboundedly.

The informal resale market is a symptomatic relief valve that makes the underlying problem (billing does not track attendance) tolerable enough that nobody — student, Mess Committee, or Warden — is ever forced to fix the fundamental cause. The workaround's success is exactly what lets the root cause persist.

B1 — Kitchen Waste Control

Balancing, but only for half the problem — a Fix That Fails.

This loop genuinely balances — it is why the toggle exists — but it only closes the kitchen-side variable. Billing (the student-side variable) has no corresponding balancing loop at all, which is precisely why B1 reads as a Fix That Fails: it looks like the no-show problem was addressed, which may reduce the perceived urgency of fixing the billing model, even though the student's actual cost problem is untouched.

Systems archetypes mapped onto current data

- **Fixes that Fail** — Skip Meal: solves kitchen waste, leaves billing broken, possibly reduces pressure to fix the real issue.
- **Shifting the Burden** — Mess Cell resale: symptomatic relief substituting for structural reform.
- **Limits to Growth** — [**Confirmed**] already happened once: Kadamba hit a capacity ceiling on non-veg demand, and Bakul was created as the workaround. [**Open question**] Has the limiting factor now shifted from kitchen capacity to student preference/trust, since Bakul has spare capacity but still is not chosen?
- **Tragedy of the Commons** — [**Hypothesis**] the shared counter/kitchen throughput during the 8:00–8:30 peak is a commons everyone draws on by timing their arrival right before class, degrading queue speed for everyone, with no individual student bearing the cost of their own contribution to it. Needs a staff perspective to confirm.
- **Overoptimization** — billing-by-registration-count optimizes for administrative simplicity and kitchen-planning predictability, at the direct expense of matching food produced to food eaten and cost to value received.
- **Ignoring stakeholders** — the Academic Office (holds the one lever most likely to help, zero engagement) and mess staff (day-to-day operational knowledge with no confirmed channel upward).
- **Manipulating/gaming the system** — the Mess Cell market itself is a mild gaming behavior: extracting value from a system not designed for resale. [**Open question**] Is this mostly accidental no-shows recovered after the fact, or do some students deliberately register broadly intending to resell what they don't use? These are different diagnoses — a design flaw producing waste, versus a rational response to a mispriced system.

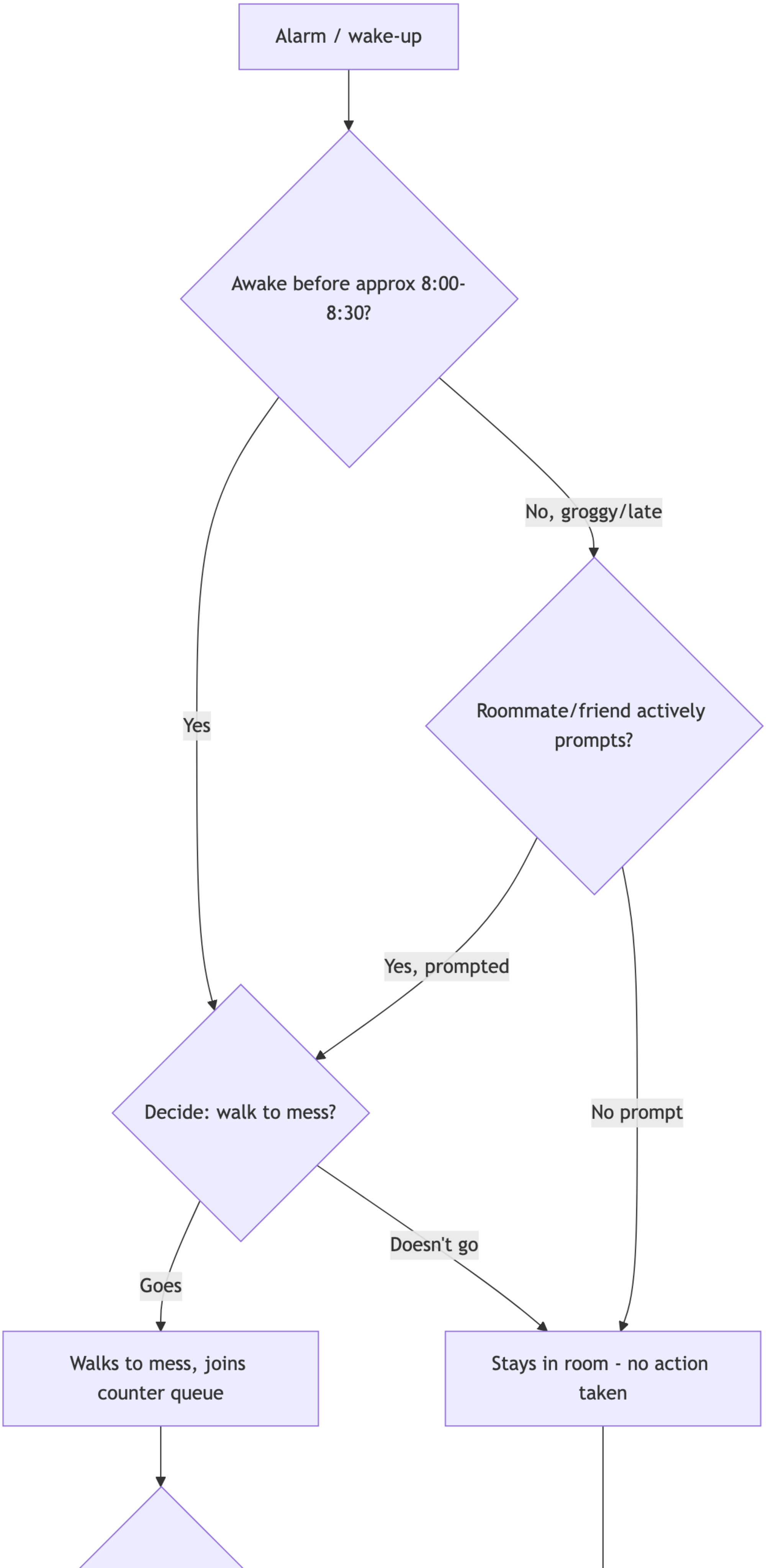
A weak/broken balancing loop worth naming

B2 — Capacity Redistribution (designed to work, doesn't appear to): the natural balancing response to Kadamba nearing capacity should be organic redistribution to Bakul/Palash. Observed data shows this is not happening — Kadamba runs near 79% of printed capacity while Bakul sits at 31%/11% (veg/non-veg) on the one date sampled. Instead of the loop self-correcting, capacity was added administratively (Bakul was built). Why the organic loop does not work — awareness, preference, or the "trusted default" mental model above — remains a real open question, not an assumption.

7 Summary and Status

#	Deliverable	Status	What closes remaining gaps
1	System Context Brief	Complete	—
2	Stakeholder Map	Complete	—
3	Stakeholder Power–Interest / Leverage Map	Draft, finalizing	Governance (Role D) interview confirms the Academic Office / vendor power reads
4	Current-State Process Map / Process Trace	Draft, pending fieldwork	Student (Role A/B) and mess-staff (Role C) interviews confirm the weekly-batching, roommate-prompt, and kitchen-side hypotheses
5	System Timeline / Behaviour-Over-Time Map	Draft, pending fieldwork	Same interviews replace the within-window hypothesis table with real recalled or observed data

Everything that could be built from existing confirmed facts and structural reasoning is done. The two remaining gaps are exactly where the framework itself requires real human input: Activity 4's "conduct interviews and observations" and Activity 5's "collect historical... observations." Both are being closed through the ongoing round of Role A (attending students), Role B (skipping students), Role C (mess operations staff), and Role D (mess governance) interviews. This report will be updated in place as each round of interview data lands.


Alarm / wake-up
Awake before approx 8:00-8:30?
Yes
No, groggy/late
Roommate/friend actively prompts?
Yes, prompted
No prompt
Decide: walk to mess?
Goes
Doesn't go
Walks to mess, joins counter queue
Stays in room - no action taken

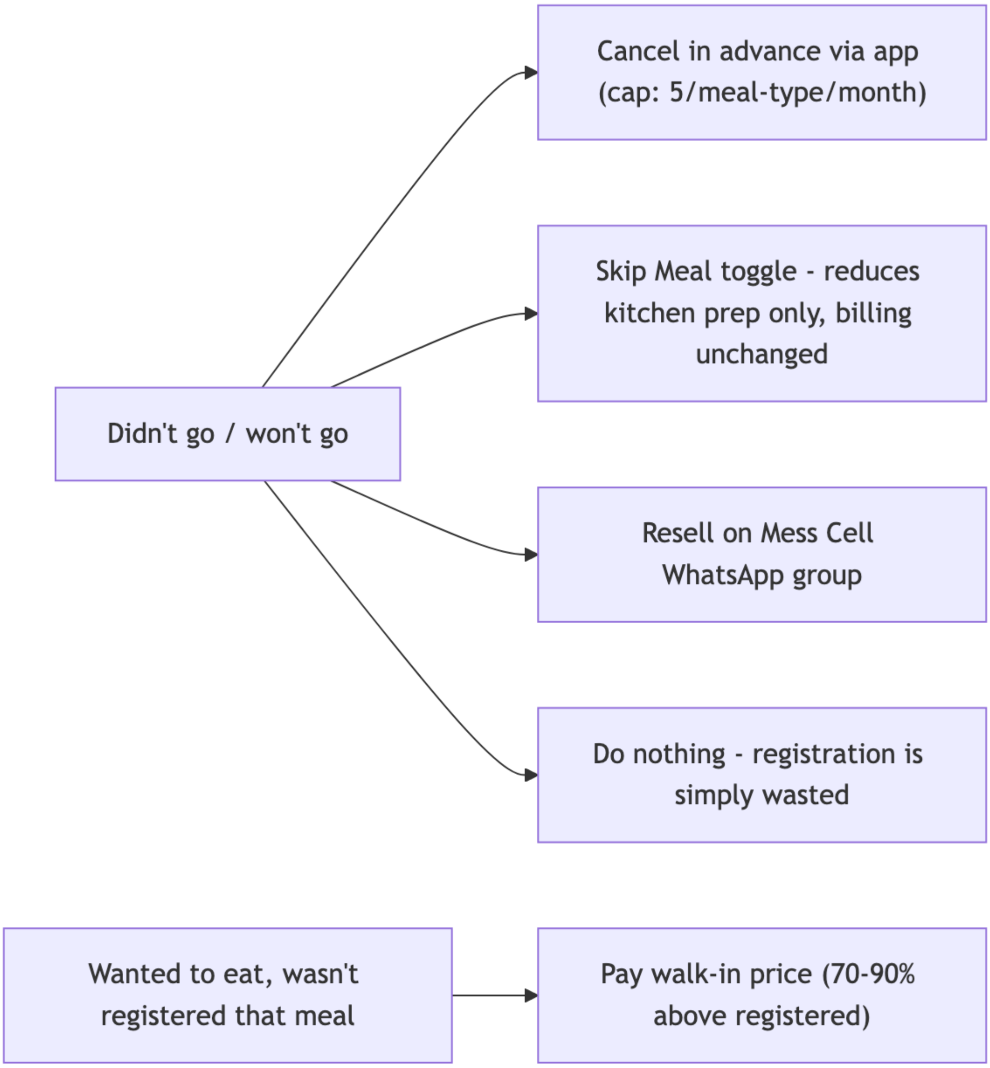


Figure 9: Exception and workaround paths branching off a no-show or an unregistered attendance attempt.

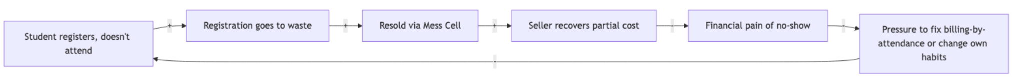


Figure 10: R1: the Mess Cell resale market as a Shifting the Burden loop.

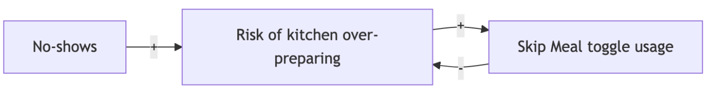


Figure 11: B1: Skip Meal balances kitchen over-preparation, but not billing.